

Selection and Heterogeneity in the Returns to Migration^{*}

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Abstract

There is considerable debate on the returns to migration and sectoral change in developing countries, and magnitudes differ sharply depending on the method used. We aim to reconcile these divergent estimates by explicitly accounting for the role of heterogeneity in the returns to migration and sectoral mobility. We exploit rich panel data from over 30 thousand individuals in Indonesia whom we observe over a 25-year period. We model self-selection into migration in a multi-period Roy model, which incorporates worker heterogeneity in both absolute and comparative advantage. We then estimate a correlated random coefficient model that considers both types of heterogeneity. This model lets us extrapolate the returns identified from switcher sub-populations to non-switchers—a group of particular interest to policymakers deciding whether to encourage migration as a development strategy. It also allows us to draw on recent developments in the literature on non-parametric panel data identification and employ a group-random coefficient estimator that explicitly tests the parametric assumptions that identify the returns to non-switchers. Our results show considerable heterogeneity in the (log income) returns to switching from agricultural to nonagricultural sectors that help explain the large variation in existing estimates. Our estimated returns to non-switchers range between 0.20 and 0.34 log points, considerably lower than the raw sectoral income gap but close to the returns to switchers identified using individual fixed effects and on par with estimates from experimental evidence.

KEYWORDS: rural-urban migration, sectoral mobility, heterogeneity, self-selection

JEL CODES: J24, J43, O13, O15, R23

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1 Introduction

As economies undergo structural transformation, labor tends to migrate out of agriculture in rural areas into higher-productivity sectors in cities. While most areas of the world have undergone rapid urbanization in the past 50 years, we still observe striking income and consumption gaps between urban and rural areas (Lagakos, 2020). Across the developing world, the average urban resident earns 2–3 times more than their rural counterparts. Large wage differentials remain even when researchers control for cost-of-living differences and a rich set of individual characteristics (Young, 2013; Gollin, Lagakos and Waugh, 2014). Ongoing debates in both academic and policy circles examine whether these earnings gaps should be interpreted as a signal that labor is misallocated across space; a corollary discussion focuses on the extent to which migration can play a role in reducing existing productivity gaps. In other words, should a social planner seek to increase the rate of rural-urban migration?

It is essential to base these debates and the accompanying policy decisions on reliable estimates of the returns to migration, meaning the causal effect of migration on earnings or consumption. Cross-sectional wage gaps, such as those in the structural transformation literature (e.g., Lagakos and Waugh, 2013) do not reflect the returns to migration because migrants differ from non-migrants in many ways, some of which are unobservable to the econometrician.

Two recent studies aim to address the issue of unobserved heterogeneity by using panel data methods that account for time-invariant individual-level characteristics. Alvarez (2020) and Hamory, Kleemans, Li and Miguel (2020) estimate the earnings gaps in data from Brazil, Indonesia, and Kenya. Depending on the specification, Hamory *et al.* (2020) find that including individual-level fixed effects reduces the earnings gaps to between 1 and 24 percent—a much lower gap than the 200–300 percent identified in cross-sectional studies.

To closer resemble a causal estimate of the returns to migration, Bryan, Chowdhury and Mobarak (2014) leverage an experiment in Bangladesh, where people randomly received an incentive to migrate in the agricultural off-season. The study estimates the average return to migration for the workers who moved in response to the experimental incentive to be roughly 30 percent.

We build on the insights from these recent studies in two main ways. First, we estimate a model of returns to migration that admits a more general form of heterogeneity than standard fixed effects models, namely a correlated random coefficient model. Our model belongs in the class of *random-coefficient* models because we allow the slope coefficients—including the returns to migration—to vary at the individual level. Within this class of models, we allow the returns to migration to be correlated with the covariates, which makes it a *correlated* random coefficient (CRC) model. In contrast, the workhorse within-estimator

often used in empirical work assumes a constant relationship between the time-varying covariates and the outcome variable (in particular, it assumes that the returns to migration are constant across workers). Further, it only allows individuals’ unobserved heterogeneity to be correlated with time-varying covariates—not with the returns to migration.

Second, our econometric framework allows us to be transparent about the assumptions under which we can infer average effects for specific subpopulations. This relates to a vast literature on microeconomic policy evaluation that examines models with heterogeneous effects and endogenous regressors. In such models, including the CRC model, it is generally challenging to extrapolate from local treatment effects, defined over some range of the regressors, to a policy-relevant parameter (Imbens, 2007). Borrowing a thought experiment from Lagakos, Marshall, Mobarak, Vernet and Waugh (2019), consider a policymaker deciding whether to allocate government resources towards reducing the costs of internal migration. Typically, she would be most interested in determining the expected effect for a randomly drawn person from the population or the average effect of the policy. In general, if the true causal returns to migration are heterogeneous, then neither fixed-effects approaches nor experimental studies recover this policy parameter.

Our approach takes advantage of the fact that our data contain different subpopulations, defined in terms of their complete migration histories. These subpopulations—which we refer to as “types”—include “switchers,” workers who move from urban to rural across periods and vice versa, as well as “non-switchers,” workers who remain in the location where we first observe them. We can consistently identify the average returns to migration for switchers under time homogeneity using the group-random coefficient model (GRC) proposed by Tjernström, Ghanem, Cabanillas, Lybbert, Michler and Michuda (2022).¹ In its simplest (unrestricted) version, the GRC is essentially a fixed-effects regression that allows for subpopulation-specific marginal effects. With additional assumptions that imply specific selection into treatment, we obtain the restricted version of the GRC and a set of testable restrictions on the relationship between a worker’s returns to migration and comparative advantage. If we do not reject the latter restriction, which Tjernström *et al.* (2022) refer to as linearity in comparative advantage (LCA), then we are able to extrapolate returns to non-switcher subpopulations.

The theoretical framework we develop in this paper is closely related to the two-period Roy (1951) models in Lagakos and Waugh (2013) and Lagakos *et al.* (2019). In the first, workers’ subsistence constraints and their sorting across sectors explain productivity gaps between the agricultural and nonagricultural sectors in different countries. In the second,

¹In our context, the time homogeneity assumption means that, conditional on the worker’s time-invariant characteristics and complete migration history, the distribution of period-specific productivity shocks does not vary over time.

workers with different migration costs and returns sort themselves across multiple regions generating different aggregate estimates for rural-urban consumption gaps. We build on these models to develop a multi-period migration model that allows for back-and-forth migration between rural and urban areas, thus matching migration trajectories observed in our data and the often-temporary nature of migration. We maintain the core assumption in Lagakos and Waugh (2013) and Lagakos *et al.* (2019) —that workers are endowed with location-specific (or sector-specific) skills—, but we modify assumptions on the role of preferences and prices in those models so that the resulting migration decision rule in our setting maps onto a correlated random coefficient model in the likes of Lemieux (1998) and Suri (2011). As a result, we can estimate heterogeneous returns to rural-urban migration and sectoral mobility between the agricultural and nonagricultural sectors using the GRC framework discussed above.

Our paper provides a way to reconcile divergent estimates in the literature by explicitly accounting for the role of heterogeneity in the returns to rural-urban migration. Different empirical methods rely on specific subpopulations for identification, and if some of those groups have higher returns than others, estimates are bound to differ. If average income gaps are partly due to efficient sorting—as posited by Lagakos and Waugh (2013), Young (2013), Hamory *et al.* (2020), and others—then existing earnings gaps would be poor predictors of the returns to migration for rural residents weighing the migration decision. Observational returns—such as those recorded by Alvarez (2020) and Hamory *et al.* (2020)—apply only to those who voluntarily move (or switch labor markets). Without further assumptions, it is hard to argue that these returns should necessarily extend to those who do not voluntarily move. In fact, returns to those who stayed behind may be smaller precisely because any higher returns have already been “arbitraged away” by the early movers.

Stayers (i.e., non-switchers) could also have high expected returns if external constraints or costs keep them from migrating in the status quo—an argument made by Lagakos, Mobarak and Waugh (2020). Experimental studies, such as Bryan *et al.* (2014), estimate the returns for individuals who are induced to move by financial incentives to migrate. Again, we have little basis for extrapolating such estimates to the broader population that did not respond to the incentives.² While these separate estimates are all informative, each approach provides but a piece of the puzzle of the returns to internal migration.

We estimate our model using data from detailed panel surveys of over 30,700 adult workers in Indonesia for whom we observe employment and location choices over 25 years. The model allows us to estimate returns to two binary dimensions for a worker’s choice: rural/urban location and agricultural/nonagricultural employment. We focus on the second

²Experimental studies and others that rely on instrumental variables approach to estimate local effects, which are not necessarily valid for the entire subpopulation of non-switchers.

because there is little heterogeneity and small returns in income to rural-urban migration in our setting but larger and more heterogeneous returns to switching from agriculture to non-agriculture. Similarly, we can estimate returns for two outcomes of interest: income and consumption. We focus on income as it provides more identifying variation in our data and better reflects worker productivity in our model (compared to consumption).

Our first set of results, obtained with traditional pooled OLS and panel regressions with individual fixed effects, corroborate findings in the literature. We find a large agricultural/nonagricultural productivity gap measured as an average difference of 67 log points in income. This gap decreases with the inclusion of different controls (to 40–42 log points) and individual fixed effects (to 24–28 log points). Because including individual fixed effects effectively restricts the identifying variation to a sample of switchers, we restrict our estimations to that sample and find a range of estimates around 30 log points. Finally, we turn to the estimation of our correlated random coefficients model, finding considerable heterogeneity in the returns to sectoral choice for different groups of switchers and estimating returns for the two groups of interest ignored in traditional estimations: those never observed in non-agriculture and those always observed in that sector.³ We find returns for the “never” type close to the average returns for switchers (20–30 log points depending on the specification used) and returns for the “always” type that are not statistically different from zero in general. We also find evidence suggesting that employment in the nonagricultural sector is a “pro-poor” technology: the trajectory types with the lowest base returns to agricultural employment get the highest incremental returns when switching to non-agriculture.

Our work contributes to two strands of the literature. First, we reconcile diverging estimates of returns to rural-urban migration and occupational mobility to the nonagricultural sector by explicitly accounting for heterogeneity in the returns of subpopulations defined by their history of location or occupation choices. We show that the returns for most of the switcher types are close to small and homogenous estimates found in the literature (Alvarez, 2020; Hamory *et al.*, 2020), whereas, in some specifications, the returns for non-switchers are close to large estimates found in the cross-sectional comparisons of the structural transformation literature (Lagakos and Waugh, 2013; Young, 2013; Gollin *et al.*, 2014).⁴ Second, we adapt existing generalized Roy models and econometric methods previously used by studies in labor (Lemieux, 1998) and technology adoption (Suri, 2011) to the study of heterogeneity in the returns to rural-urban migration. Our empirical strategy builds on recent work in

³By “ignored”, we mean that these groups do not contribute to the identifying variation in estimations with individual fixed effects.

⁴Lagakos *et al.* (2020) also reconcile estimates, but their focus is on reconciling early experimental evidence of high returns to temporary rural-to-urban migration with later experiments that showed much smaller estimates.

the econometrics literature (Tjernström *et al.*, 2022; Verdier, 2020) to estimate returns to switcher subpopulations in a flexible manner while also assessing the validity of the identifying restrictions necessary to estimate returns to non-switchers. Our model can be extended to accommodate a richer discussion of the role of temporary and permanent costs in the worker’s migration decisions, and our empirical strategy can be used by studies that aim to reconcile estimates from analysis with cross-sectional, experimental and longitudinal designs.

2 Model

We consider an economy with two sectors, agriculture (A) and non-agriculture (N), into which individuals (i) sort based on their skills and preferences.⁵ As is common in the literature (Bazzi, Gaduh, Rothenberg and Wong, 2016; Alvarez, 2020; Lagakos and Waugh, 2013; Pulido and Świącki, 2021), we adapt the classic Roy (1951) model to motivate our empirical strategy. In line with this literature, we allow for workers’ unobservable comparative advantage to play a role in their choice to migrate or switch sectors.

The different sectors in the economy require different skills from the workers they employ. Workers in the economy have different endowments, summarized by a vector of sector-specific skills $a_i = \{a_i^A, a_i^N\}$. These skills determine a worker’s productivity when carrying out sector-specific tasks. Each person in the economy receives a draw from this joint distribution of skills, $G(a^A, a^N)$, which we assume has support on the positive reals and finite first two central moments.

2.1 Income, utility, and sectoral choices

We assume that the log income of individual i in period t is determined by her unobserved ability, a_i , her sectoral choice in that period, D_{it} , her vector of observed characteristics, X_{it} , and an error term, ε_{it} . By assuming that the last two components are additive, and omitting the vector of observed characteristics for now, individual i ’s log income, for $i = 1, \dots, n$ and $t = 1, \dots, T$ is given by:

$$y_{it} = f(a_i, D_{it}) + \varepsilon_{it}. \quad (1)$$

Let $D_{it} = 1$ if the individual works in non-agriculture and 0 otherwise. The relationship between unobserved ability and sectoral choice given by f takes only two possible values per individual. The first, $f(a_i, 0)$, is the individual-specific log income when

⁵Throughout this section, we discuss sectoral choice, in line with Roy (1951) model. The implications extend naturally to migration decisions.

choosing the agricultural sector. The second, $f(a_i, 1)$, is her log income when choosing the nonagricultural sector. Denoting the first value by $\mu_i \equiv f(a_i, 0)$ and their difference by $\Delta_i \equiv f(a_i, 1) - f(a_i, 0)$, we can write an individual's log income in the following intercept-and-slope format:

$$y_{it} = \mu_i + \Delta_i D_{it} + \varepsilon_{it}. \quad (2)$$

In this scenario, and ignoring the error term ε_{it} for now, an individual chooses the nonagricultural sector whenever the income premium associated with such choice is positive:

$$D_{it} = 1 \Leftrightarrow (y_{it}|a_i, D_{it} = 1) - (y_{it}|a_i, D_{it} = 0) = \Delta_i > 0. \quad (3)$$

By assuming that unobserved ability is time-invariant, we assume time-homogeneity in the individual-specific premium, Δ_i . In this context, an individual would only switch sectors after the first period if (i) there is a change in her observed characteristics that are rewarded differently in each sector (e.g., schooling), or (ii) there are fluctuations in sector-specific factors that affect all individuals equally (e.g., firms' productivity).⁶

We assume that individuals make their sectoral decisions based not on income alone but on overall utility, and this utility is subjected to idiosyncratic shocks that vary by sector and time. Though income surely has a large weight in an individual's utility function, it is not the only argument. Individuals also have preferences over non-monetary things such as proximity to family, local amenities, and other factors that vary by sector and time. We allow such factors to affect one's sectoral decisions by introducing a simple utility shock. With this modification, individuals in our model may switch sectors even when their returns are time-invariant, and there are no differences in the rewards to observed characteristics or sector-specific factors that affect all individuals equally.

Specifically, we let ν_{it}^s denote the individual i 's idiosyncratic utility shock when she is in sector $s = \{\text{Agriculture, Non-Agriculture}\}$, and $U_{it}^s = y_{it} + \nu_{it}^s$ denote her utility in period t . At the end of each period, this individual realizes her income, observes her utility shocks, and forms expectations on the shocks for the next period. She then decides to switch to (or remain at) the nonagricultural sector at the beginning of period t if her expected utility there is higher. That is, she chooses $D_{it} = 1$ if

$$\mathbb{E}(U_{it}^N) > \mathbb{E}(U_{it}^A) \Leftrightarrow \Delta_i + \mathbb{E}(\nu_{it}^N - \nu_{it}^A) > 0. \quad (4)$$

In what follows, we assume that individuals make their sectoral choices in each period

⁶In the second case, the order (in terms of ability) of the individuals crossing the decision threshold would always be the same: a positive fluctuation that moves someone with Δ_i over the threshold will necessarily move all other individuals i' with $\Delta_{i'} > \Delta_i$.

according to the decision rule in equation (4). Implicit in this equation, is the assumption that the individual knows her ability and the returns to such ability in both sectors ($\mu_i + \Delta_i$) and, in the model with observed characteristics and time fixed effects, δ_t , the individual also knows the returns in each sector to her characteristics ($X'_{it}(\gamma^N - \gamma^A)$), and any difference in time fixed effects ($\delta_t^N - \delta_t^A$). Finally, equation (4) implies that, for the individual: $\mathbb{E}(\varepsilon_{it}|a_i, D_{it} = 1) - \mathbb{E}(\varepsilon_{it}|a_i, D_{it} = 0) = 0$.

2.2 Heterogeneous returns for different trajectory types

Settings like ours, with a binary choice and multiple periods, allow for grouping individuals into different types based on their full history of sectoral choices, i.e., their trajectories of nonagricultural employment. For example, with two periods, we can group individuals into four types: those never observed in the nonagricultural sector, those who start in agriculture but join the nonagricultural sector in the second period, those who start in non-agriculture but leave for the agricultural sector in the second period, and those always observed in non-agriculture. Since an individual's trajectory in the two-period case is given by $D_i = \{D_{i1}, D_{i2}\}$ and our convention is that $D_{it} = 1$ if non-agriculture and 0 otherwise, these four types can be written concisely as $\{0, 0\}$, $\{0, 1\}$, $\{1, 0\}$ and $\{1, 1\}$. In the general case with T periods, an individual's trajectory $D_i = \{D_{i1}, \dots, D_{iT}\}$ will assign her to one of 2^T possible types. The first and last of these types are composed of individuals who never switch sectors, those for whom D_{it} is always equal to zero or always equal to one. Everyone else, those with $D_{it} \neq D_{it'}$ for some $t \neq t'$, is in a switcher type.

Trajectory types represent a relevant dimension of heterogeneity in the returns to sectoral mobility. Traditional panel estimations with individual fixed effects estimate a unique average return that is a weighted average of the returns for all switcher types. The non-switchers provide no identifying variation. Such averaging hides heterogeneity. For example, the nonagricultural premium for the $\{0, 1\}$ type can be larger than the agricultural penalty for the $\{1, 0\}$ type. Individuals may gain 20% more on average after moving from agriculture to non-agriculture but lose 30% on average after switching sectors in the opposite direction. In other words, the changes in income associated with a change in sector for these two switcher types can have opposite directions, which is intuitive, but their magnitudes may not be symmetric.⁷ More importantly, traditional panel estimations say nothing about the returns for the non-switcher types. We cannot estimate the returns for non-switchers unless we assume these returns are equal to some weighted average of the returns for switchers, which is unlikely to be true.

⁷Pulido and Świącki (2021) show evidence of such an asymmetry in the returns to sectoral mobility in Indonesia (see their Table 4).

Heterogeneity in the returns for different trajectory types does not preclude heterogeneity in other dimensions such as gender, or age group. In fact, all these different heterogeneities and their intersections can be investigated in empirical studies. Nonetheless, focusing on trajectories as the main dimension of heterogeneity in the returns to sectoral mobility has advantages. First, trajectories can be related to individual unobserved ability. For example, if sorting is efficient, individuals with the lowest returns to non-agriculture will seldom be observed in that sector (the opposite will be true for those with high returns). And individuals moving back-and-forth between agriculture and non-agriculture will be those for whom the returns to both sectors are similar. Also, trajectory groups—especially when split into two bigger groups, switchers and non-switchers—overlap with groups of interest in public policy interventions that can specifically target marginal or infra-marginal movers. Finally, the full trajectory of nonagricultural employment across all T periods in the data, albeit unknown to the individual at every period t except the last, is known to the econometrician who can leverage this information for identification purposes. In fact, this is what we do in our empirical strategy, following other papers in the correlated random coefficients literature like Lemieux (1998) and Suri (2011).⁸

We can express the expected value of log income, $\mathbb{E}(y_{it})$, as a combination of different group averages. Let $d \in \mathcal{D} = \{0, 1\}^T$ denote any possible realized trajectory for T periods. Since the set of all possible trajectories $\mathcal{D}$ is finite, under the strict exogeneity assumption, $\mathbb{E}(\varepsilon_{it}|a_i, D_i) = 0$, we can integrate log income across all the 2^T possible trajectory groups in the data obtaining

$$\mathbb{E}(y_{it}|D_i = d) = \mu_d + \Delta_d D_{it}, \quad (5)$$

where $\mu_d = \mathbb{E}(\mu_i|D_i = d)$ is the group-average return to choosing agriculture, and $\Delta_d = E(\Delta_i|D_i = d)$ is the group-average nonagricultural premium, i.e., the incremental return to choosing non-agriculture.

Equation (5) can be readily estimated in our data using a regression specification that contains a different dummy variable for each trajectory type, interactions of the nonagricultural dummy and the dummies for all switcher types, and additive control variables like observed individual characteristics and time fixed effects. This estimation identifies heterogeneous returns to choosing the nonagricultural sector for all trajectory types except two: the non-switchers. For those who never choose non-agriculture ($d = \{0, \dots, 0\}$) we can only identify μ . And for those who always choose non-agriculture ($d = \{1, \dots, 1\}$) we can only

⁸Specifically, we assume strict exogeneity in the error term, $\mathbb{E}(\varepsilon_{it}|a_i, D_i) = 0$, which conditions the expectation of the error term on the entire trajectory of sectoral choices of an individual (past, present, and future). In a model with observed characteristics, X_{it} , and time fixed effects, δ_t , the strict exogeneity assumption changes to $\mathbb{E}(\varepsilon_{it}|a_i, X_{it}, \delta_t, D_i) = 0$.

identify a combined parameter $\kappa = \mu + \Delta$. To identify Δ for the first type and to separately identify μ and Δ for the last we must impose additional restrictions to our model.

2.3 Decomposing unobserved ability into absolute and comparative advantage

We assume unobserved ability can be decomposed into three parts or sets of skills: one that is equally useful (and rewarded) in both sectors, τ_i , one that is more useful in agriculture, θ_i^A , and one that is more useful in non-agriculture, θ_i^N . The difference between the θ_i^N and θ_i^A skill sets can be interpreted as the individual's comparative advantage in non-agriculture, and the τ_i skill set as her absolute advantage. By assuming that unobserved ability is time-invariant, we are assuming that an individual is born with such a tripartite skill set.⁹

Next, like Lemieux (1998) and Suri (2011), we impose a specific structure to the relationship between these skill sets, redefining their relationship as linear projections:

$$\theta_i^A = b_A (\theta_i^N - \theta_i^A) + \tau_i \text{ and } \theta_i^N = b_N (\theta_i^N - \theta_i^A) + \tau_i, \quad (6)$$

where by defining $\sigma_N^2 \equiv \text{Var}(\theta_i^N)$, $\sigma_A^2 \equiv \text{Var}(\theta_i^A)$, and $\sigma_{NA} \equiv \text{Cov}(\theta_i^N, \theta_i^A)$ we have $b_N = (\sigma_N^2 - \sigma_{NA}) / (\sigma_N^2 + \sigma_A^2 - 2\sigma_{NA})$ and $b_A = (\sigma_{NA} - \sigma_A^2) / (\sigma_N^2 + \sigma_A^2 - 2\sigma_{NA})$.

Within this structure, the term τ_i does not vary with the individual's sector and is orthogonal to the difference $\theta_i^N - \theta_i^A$ by construction. Defining $\theta_i \equiv b_A (\theta_i^N - \theta_i^A)$ and $\phi \equiv (b_N - b_A) / b_A$, we can rewrite both sector-specific skill sets as functions of the individual's comparative advantage θ_i and the parameter ϕ :

$$\theta_i^A = \theta_i + \tau_i \text{ and } \theta_i^N = (1 + \phi) \theta_i + \tau_i. \quad (7)$$

Finally, we assume that absolute advantage is mean independent of the sectoral choice, $\mathbb{E}(\tau_i | D_i) = \mathbb{E}(\tau_i)$, we normalize the comparative advantage to be mean zero, $\mathbb{E}(\theta_i) = 0$, and we let the return to non-agriculture have a component that is common to all individuals, β .¹⁰ With these additional assumptions and normalization, we obtain a restricted version of equation (2):

$$y_{it} = \theta_i + \tau_i + (\beta + \phi\theta_i) D_{it} + \varepsilon_{it}, \quad (8)$$

⁹The assumption that individuals are endowed with a vector of market-specific skills or abilities is common in the literature on rural/urban productivity gap (or agricultural/nonagricultural, like here). See, for example, Lagakos and Waugh (2013), Lagakos *et al.* (2020), and Pulido and Świącki (2021).

¹⁰These are the same assumptions made by Suri (2011), which in turn follows Lemieux (1993, 1998) and Carneiro, Hansen and Heckman (2003).

where $\mu_i = \theta_i + \tau_i$ and $\Delta_i = \beta + \phi\theta_i$.

2.4 Interpreting the linear extrapolation that identifies returns to non-switchers

Equation (8) says that an individual's return to choosing agriculture, μ_i , is determined by her unobserved ability in that sector (from equation (7), $\theta_i^A = \theta_i + \tau_i$), whereas her incremental return to choosing non-agriculture, Δ_i , is determined by a non-agriculture premium that is common to all individuals, β , and the parameter ϕ . This parameter directly relates to the individual's comparative advantage in the nonagricultural sector. In fact, equation (7) implies that the ratio of an individual's sector-specific skills net of her absolute advantage, i.e., her net comparative advantage in the nonagricultural sector is equal to $1 + \phi$, which in turn represents a relationship between the dispersion of these skills and their correlation in the population:

$$\frac{\theta_i^N - \tau_i}{\theta_i^A - \tau_i} = 1 + \phi = \frac{\text{Var}(\theta_i^N) - \text{Cov}(\theta_i^N, \theta_i^A)}{\text{Cov}(\theta_i^N, \theta_i^A) - \text{Var}(\theta_i^A)} \quad (9)$$

Equation (9) has two important takeaways. The first is the interpretation of comparative advantages in terms of the variance and covariance of sector-specific skills in the population, as one would expect in any generalized Roy (1951) model. The second takeaway is a direct result from the linear projections used in equation (6), which yields a relationship between the individual's pair of sector-specific skills that is also linear. Such relationship is key to the identification of the returns to non-switcher types in our model, which rely on a linear extrapolation from the returns to switchers (usefully illustrated by Verdier (2020)). Tjernström *et al.* (2022) call this the Linearity in Comparative Advantage (LCA) assumption and show that for any two trajectory types $d \neq d'$ the following equality holds:¹¹

$$\phi = \frac{\Delta_d - \Delta_{d'}}{\mu_d - \mu_{d'}}, \quad \mu_d \neq \mu_{d'}. \quad (10)$$

A particularly useful way to interpret the linear extrapolation in equation (10) is to think of employment in non-agriculture as an income enhancing technology, and the corresponding sectoral choice as technology adoption. Then, a positive ϕ means the technology gives higher returns for those who start ahead (it is “pro-rich”): for any two trajectory types, the type with the highest average return to choosing agriculture will also have the highest average incremental return to choosing non-agriculture. Conversely, a negative ϕ means the technology is “pro-poor:” it is the type with the lowest average return in agriculture who

¹¹See proposition 1 in their paper.

will benefit the most from switching to non-agriculture.

The restricted model in equation (8) can be estimated using the flexible Group Correlated Coefficient (GRC) approach proposed by Tjernström *et al.* (2022). Under strict exogeneity of the error term and the LCA assumptions, the GRC identifies returns to non-switchers. It also provides a framework in which we can test key identifying restrictions and calculate weak-identification robust inference confidence intervals for the parameters of interest.

Figure 1 below illustrates the linear extrapolation determined by equation (10) in a two-period panel, in which the restricted GRC model is exactly identified. The figure shows estimates for the base level return μ and the income increment Δ for the two switcher types $d = \{0, 1\}$ and $d = \{1, 0\}$ on the (μ, Δ) -plane. The vertical line denotes the base level for the non-switcher type $d = \{0, 0\}$, whereas the 45-degree line denotes the combination of base level and increment for the non-switcher type $d = \{1, 1\}$, i.e., the combined parameter $\kappa = \mu_{\{1,1\}} + \Delta_{\{1,1\}}$. All these parameters can be estimated using the unrestricted version of the GRC (equation (5)). The restricted GRC imposes a common intercept ϕ for the line that connects the (μ, Δ) -estimates for all types, both switchers and non-switchers. From the switchers, we obtain the extrapolation line. And from the intersection of this line with the vertical and 45-degree lines, we obtain estimates for the non-switcher types.

3 Data

3.1 Source

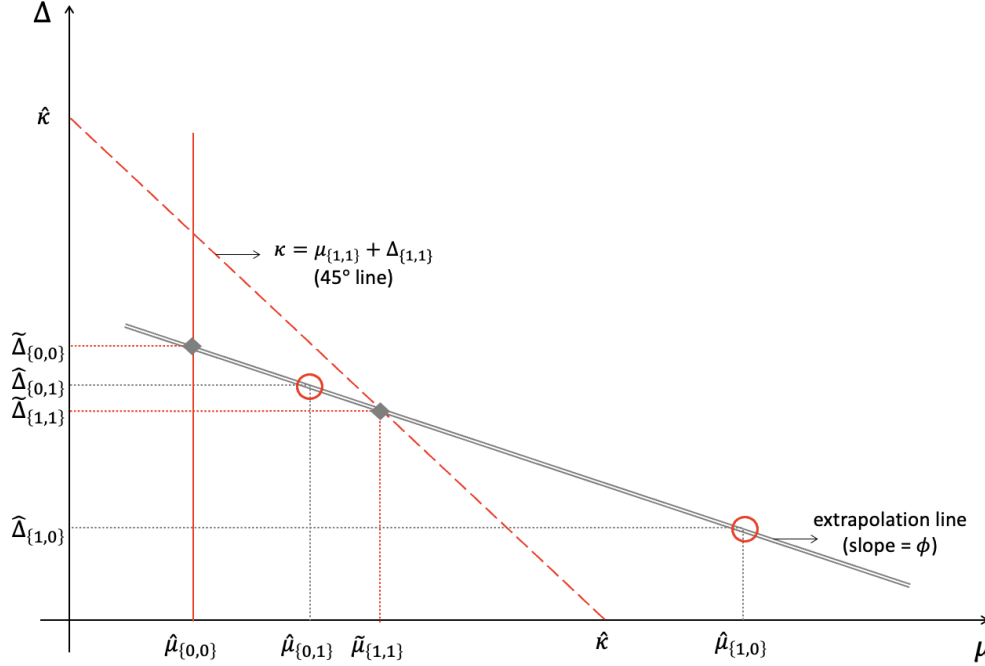
We use the Indonesia Family Life Survey (IFLS) to test the predictions of the model described in the previous section. The IFLS is a panel dataset that is known for relatively low rates of attrition and is representative of about 83% of the Indonesian population (Strauss *et al.*, 2004)¹². The analyses are based on all five waves of the IFLS, collected between 1993 and 2015 (Strauss *et al.*, 2016).

The original 1993 sample consisted of 22,347 individuals, but with the inclusion of additional members from split-off households in subsequent rounds a sample of 58,337 unique individuals is obtained. Detailed data on current and previous employment and location were collected during each IFLS survey round, allowing us to create up to a 28-year annual individual employment and migration panel, from 1988 to 2015.¹³

¹²Attrition is often high in panel data, but with an intensive focus on respondent tracking, the IFLS is well-suited to study migration. Re-contact rates between any two rounds are above 90%, and 87% of original households were contacted in all five rounds (Strauss, Witoelar and Sikoki, 2016). Thomas *et al.* (2012) contains a detailed discussion of tracking and attrition in the IFLS.

¹³Refer to Kleemans (2022) and Kleemans and Magruder (2018) for more details on the IFLS panel dataset.

Figure 1: The extrapolation line with two periods



Notes: The parameter $\mu = \mathbb{E}(y_{it}|D_{it} = 0)$ in the x-axis denotes the base income level in the agricultural sector, and the parameter $\Delta = \mathbb{E}(y_{it}|D_{it} = 1) - \mathbb{E}(y_{it}|D_{it} = 0)$ in the y-axis denotes the increment in income received in the nonagricultural sector over the base level. The subscripts in each estimate denote the type $d \in \{0, 1\}^T$. We place a “hat” over the estimates from the unrestricted version of the GRC: μ s and Δ s for the two switcher types $\{0, 1\}$ and $\{1, 0\}$, μ for the non-switcher type $\{0, 0\}$, and $\kappa = \mu + \Delta$ for the non-switcher type $\{1, 1\}$. We place a “tilde” over the estimates identified by the extrapolation line in the restricted version of the GRC: Δ for the non-switcher type $\{0, 0\}$, and μ and Δ for the non-switcher type $\{1, 1\}$.

The IFLS consists of 5 waves conducted in 1993, 1997/1998, 2000, 2007/2008, and 2014/2015, and each wave collects detailed information on individuals and their households. In this study, we use data on individuals’ location (urban or rural), sector of employment (agriculture or non-agriculture), income, and demographic information, such as gender, age, and education. We keep all individuals aged 16 and above with non-missing information on total income, sector of main work, and location. The resulting sample has 66,325 observations of 30,759 individuals, of which 18,614 appear at least twice in the data. Of those individuals with at least two observations, 9,611 switch sectors at least once, and 1,418 individuals appear in all five waves.

We organize our data into three different samples, for which we present descriptive statistics and results in the following sections. First, we consider all observations in our selection of the IFLS data. This sample includes both individuals who switch and do not switch sectors as well as individuals who appear only once in the data. Second, we restrict our sample to switchers. This sample keeps only the individuals who switch between agricultural and nonagricultural employment at least once. Third, we impose perfect balance in the data

restricting it to individuals who appear in all five waves of the IFLS. This sample includes both switchers and non-switchers.

3.2 Descriptive statistics

Table 1 shows how sectoral (Agriculture and Non-Agriculture) and location (Rural and Urban) choices overlap in our data for all three samples. We note a similarity between the Full and Balanced samples. In both, the majority of individuals (70%) work in non-agriculture, but slightly less than half live in urban locations. These proportions are somehow inverted for the Switchers sample: about half work in non-agriculture, and less than 30% live in urban locations. Switchers differ in important ways from individuals in the Full sample, and these differences remain when we impose a perfect balance on our panel data. We also note that while there is a substantial share of people in rural areas that work in non-agriculture (between 29% and 34% across the three samples), the opposite is not true: only 5% of the individuals in urban locations in both the Full and Balanced samples work in agriculture. (This share increases to 12% in the Switchers sample.) There is a strong correlation between urban location and nonagricultural work, but not between rural location and agricultural work.

Table 1: Overlapping of sectoral and location choices

Sector or Location	Full sample		Switchers sample		Balanced sample	
	obs.	share	obs.	share	obs.	share
Agriculture	18,578	28%	4,783	50%	2,313	33%
Rural	15,426	23%	3,633	38%	1,932	27%
Urban	3,152	5%	1,150	12%	381	5%
Non-Agriculture	47,747	72%	4,828	50%	4,777	67%
Rural	19,068	29%	3,297	34%	2,056	29%
Urban	28,679	43%	1,531	16%	2,721	38%
Total Rural	34,494	52%	6,930	72%	3,988	56%
Total Urban	31,831	48%	2,681	28%	3,102	44%
Total	66,325	100%	9,611	100%	7,090	100%

It is worth mentioning some figures not shown in the table, namely, the number and proportion of individuals in some trajectory types. In the Balanced sample, we have 279 individuals (20%) who are never observed in the nonagricultural sector (the “never”

type) and 768 individuals (54%) who are always observed in the nonagricultural sector (the “always” type). The remaining 371 individuals (26%) are observed switching sectors at least once and are thus distributed among the switcher types.

In Table 2, we show some descriptive statistics and differences in means for our three study samples. Our focus here, as in the main analysis and results, is on differences across the agriculture and non-agriculture sectors. Once more, the Full and Balanced samples show similar patterns. On average, the individuals working in agriculture have lower incomes, are less likely to be born in urban locations and be female, live in smaller households, and are older and less educated than individuals working in non-agriculture. For the sample of switchers, the differences are much less significant, both in the statistical and economic senses. The only notable difference remaining is for log income. Switchers working in agriculture have lower income, but the difference in this sample is around half as large as it is for the Full and Balanced samples (-0.32 log points versus -0.67 and -0.68).

4 Results

4.1 Results from OLS estimations

We start by presenting OLS estimates for the income premium associated with nonagricultural employment. The results show how the difference in average income between those employed in agriculture and non-agriculture changes as we move across specifications that control first for observed characteristics, time fixed effects, and location (province) fixed effects and later for individual fixed effects as well. By varying specifications, we aim to elicit the underlying determinants of the sectoral income premium. In particular, we want to shed light on the role of skill differences of the individuals who sort themselves in either agriculture or non-agriculture.

We repeat this analysis for three different samples: all individuals in our selection of the IFLS data, only those who switch between agricultural and nonagricultural employment at least once, and only individuals who appear in all five waves of the IFLS. By varying our analysis between these samples, we accomplish two goals. First, we investigate if it is the unobserved ability or sample restrictions that drive the reduction in coefficients observed when individual-fixed effects are added to the regression specifications. Second, we provide comparability between the OLS results and the GRC results shown later, which require a perfectly balanced sample.

Table 3 below presents our OLS estimates. The dependent variable in all regressions is the log of total income. The top panel shows results for the Full sample, the middle panel shows results for the Switchers sample, and the bottom shows results for the Balanced sample.

Table 2: Summary statistics and differences in means

Variable	All Observations		Agriculture	Non-Agric.	Difference (A – N)	
	mean	s.d.	mean	mean	mean	p-value
<i>Panel A: Full sample, N = 66,325 (30,759 Individuals)</i>						
Non-Agriculture	0.72	0.45	0	1	.	.
Log Ingome	14.90	1.14	14.42	15.09	-0.67	0.00
Born Urban	0.28	0.45	0.10	0.35	-0.25	0.00
Female	0.39	0.49	0.30	0.43	-0.13	0.00
HH Size (members)	4.77	2.19	4.65	4.82	-0.17	0.00
Age (years)	38.98	13.40	43.82	37.10	6.72	0.00
Education (years)	7.99	4.59	5.30	9.04	-3.74	0.00
<i>Panel B: Switchers sample, N = 9,611 (2,915 Individuals)</i>						
Non-Agriculture	0.50	0.50	0	1	.	.
Log Ingome	14.71	1.12	14.55	14.87	-0.32	0.00
Born Urban	0.14	0.34	0.13	0.15	-0.02	0.00
Female	0.28	0.45	0.27	0.28	-0.01	0.15
HH Size (members)	4.76	2.08	4.77	4.76	0.01	0.86
Age (years)	41.56	12.94	41.71	41.40	0.30	0.25
Education (years)	6.31	4.06	6.14	6.49	-0.35	0.00
<i>Panel C: Balanced sample, N = 7,090 (1,418 Individuals)</i>						
Non-Agriculture	0.67	0.47	0	1	.	.
Log Ingome	15.05	1.05	14.60	15.28	-0.68	0.00
Born Urban	0.23	0.42	0.11	0.29	-0.18	0.00
Female	0.28	0.45	0.17	0.34	-0.17	0.00
HH Size (members)	4.84	2.03	4.68	4.92	-0.24	0.00
Age (years)	43.78	11.21	46.32	42.55	3.78	0.00
Education (years)	7.55	4.45	5.57	8.50	-2.93	0.00

We report the number of observations and individuals in each sample and keep regression specifications consistent across columns. The main explanatory variable is a dummy equal to one when the individual's main work is in the nonagricultural sector. We do not report coefficients for the intercept or any other covariates.

Table 3: OLS estimates of the returns in log income from nonagricultural employment

Dependent variable:	Log income						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A: Full sample, N = 66,325 (30,759 Individuals)</i>							
NonAg. Employment	0.6658 (0.0098)***	0.4215 (0.0099)***	0.4158 (0.0098)***	0.3985 (0.0099)***	0.2775 (0.0210)***	0.2499 (0.0208)***	0.2422 (0.0206)***
<i>Panel B: Switchers sample, N = 9,611 (2,915 Individuals)</i>							
NonAg. Employment	0.3229 (0.0227)***	0.2990 (0.0207)***	0.2796 (0.0207)***	0.2766 (0.0206)***	0.2775 (0.0210)***	0.2539 (0.0208)***	0.2438 (0.0207)***
<i>Panel C: Balanced sample, N = 7,090 (1,418 Individuals)</i>							
NonAg. Employment	0.6784 (0.0262)***	0.4910 (0.0272)***	0.4963 (0.0270)***	0.5296 (0.0281)***	0.2997 (0.0486)***	0.3175 (0.0493)***	0.3166 (0.0492)***
Baseline Covariates		Y	Y	Y			
Time (wave) FE			Y	Y		Y	Y
Province FE				Y			Y
Individual FE					Y	Y	Y

Notes: The Switchers sample is restricted to individuals who change their agricultural employment status at least once. The Balanced sample is restricted to individuals who appear in all five waves of the IFLS data. Baseline controls are an indicator for born urban, female dummy, household size (nr. of members), age and its square, and years of education and its square, all measured in the baseline (first wave). nonagricultural Employment is an indicator equal to one if the individual's main job is in the nonagricultural sector. Robust standard errors in parentheses. Stars denote: * p<0.10; ** p<0.05; *** p<0.01.

In column one, the specification yields the raw log income gap between the agricultural and nonagricultural sectors, which amounts to a difference in means. In column two, we add baseline covariates, the value of individual characteristics observed in the first period. The characteristics included are an indicator for being born in an urban location, a female dummy, the number of members in the household, age and its square, and years of education and its square. In column three, we add time (IFLS wave) fixed effects, and in column four, we add location (province) fixed effects.¹⁴ Finally, we add individual fixed effects to control for time-invariant unobserved ability in columns five, six, and seven.

¹⁴There is little change in the coefficients if we use time-varying individual characteristics or lower administrative levels for the location fixed effects. Also, because the baseline covariates are time-invariant by construction, they drop when individual fixed effects are added.

The estimates in column one of panel A corroborate a claim often made in the literature: They show that the agricultural/nonagricultural productivity gap, as measured by the difference in average income, is huge. Employment in the nonagricultural sector is associated with an income premium of 67 log points [lp] on average (or 95%).¹⁵ The premium drops to 42 lp when baseline covariates are added in column two, and it remains close to this value in columns three and four when time and province fixed effects are added. The income premium associated with nonagricultural employment drops further to the 24–28 lp range in columns five to seven, in which we add individual fixed effects to the regression specifications.

The decrease in the magnitude of the nonagricultural premium when individual fixed effects are added could be due to a significant portion of the nonagricultural premium coming from the selection of more productive people in agriculture. Thus, the nonagricultural premium decreases once we control for unobserved time-invariant ability. Such an explanation is frequent in the literature and could fit as an explanation for our results as well. However, we note that the results reported in columns five to seven come from estimations that effectively disregard, for the purpose of estimating the nonagricultural premium, all individuals who never switch sectors. While it is possible that the nonagricultural premium in columns one to four is inflated by unobserved ability, it is also possible that the nonagricultural premium for the individuals that did not contribute to the estimation of the premium in columns five to seven (the non-switchers), is indeed larger than the premium of switchers. We cannot measure such heterogeneity using traditional panel estimations with individual fixed effects.

The results in panel B, in which we restrict the sample to switchers only, suggest that it is not so much the inclusion of individual-fixed effects but the effective sample restrictions that drive the reduction in the coefficients for the nonagricultural premium. The range of coefficient magnitudes in columns two to four (30–28 lp) in panel B is quite close to the range in columns five to seven (28–24 lp). Even the coefficient in column one, 32 lp, which reflects a simple difference in means between the two sectors for the Switchers sample, is close to the range of magnitudes we obtain when individual fixed effects are added to the regression specifications for both the Full and the Switchers samples. These results indicate that the sample selection may be driving results towards lower magnitudes more than the inclusion of individual fixed effects. In other words, these results support the idea that switchers and non-switchers represent groups with distinct returns to sectoral changes. They also underpin the importance of estimating returns to non-switchers.

In panel C, we look at estimates for the Balanced sample, the same sample we use for

¹⁵Our coefficients are all fairly large in magnitude and statistically significant at the 1% level or lower. Like Pulido and Świącki (2021), we discuss coefficients using log points without mentioning their statistical significance and only occasionally calculating their corresponding percentage differences.

restricted GRC estimations that yield estimates for returns to nonagricultural employment for non-switchers. Repeating the OLS estimations for this sample allows us to check how they differ from the results for the Full sample. It also allows us to compare OLS and GRC estimates holding the sample unchanged. The pattern of the results for the Balanced sample is similar to that of the Full sample. We observe similar changes when moving across specifications, but the coefficient magnitudes are all larger, sometimes by a factor of 0.3. This increase in magnitudes may be due to the selection of individuals into the perfectly balanced sample. Individuals in the Balanced sample reported earnings in all five rounds of the IFLS, while those in the Full sample either dropped out of the IFLS data or did not report income in at least one wave. Individual characteristics associated with the probability of holding an income-generating activity in five different years and reporting it to the IFLS surveyors may also be associated with having higher returns to nonagricultural employment.

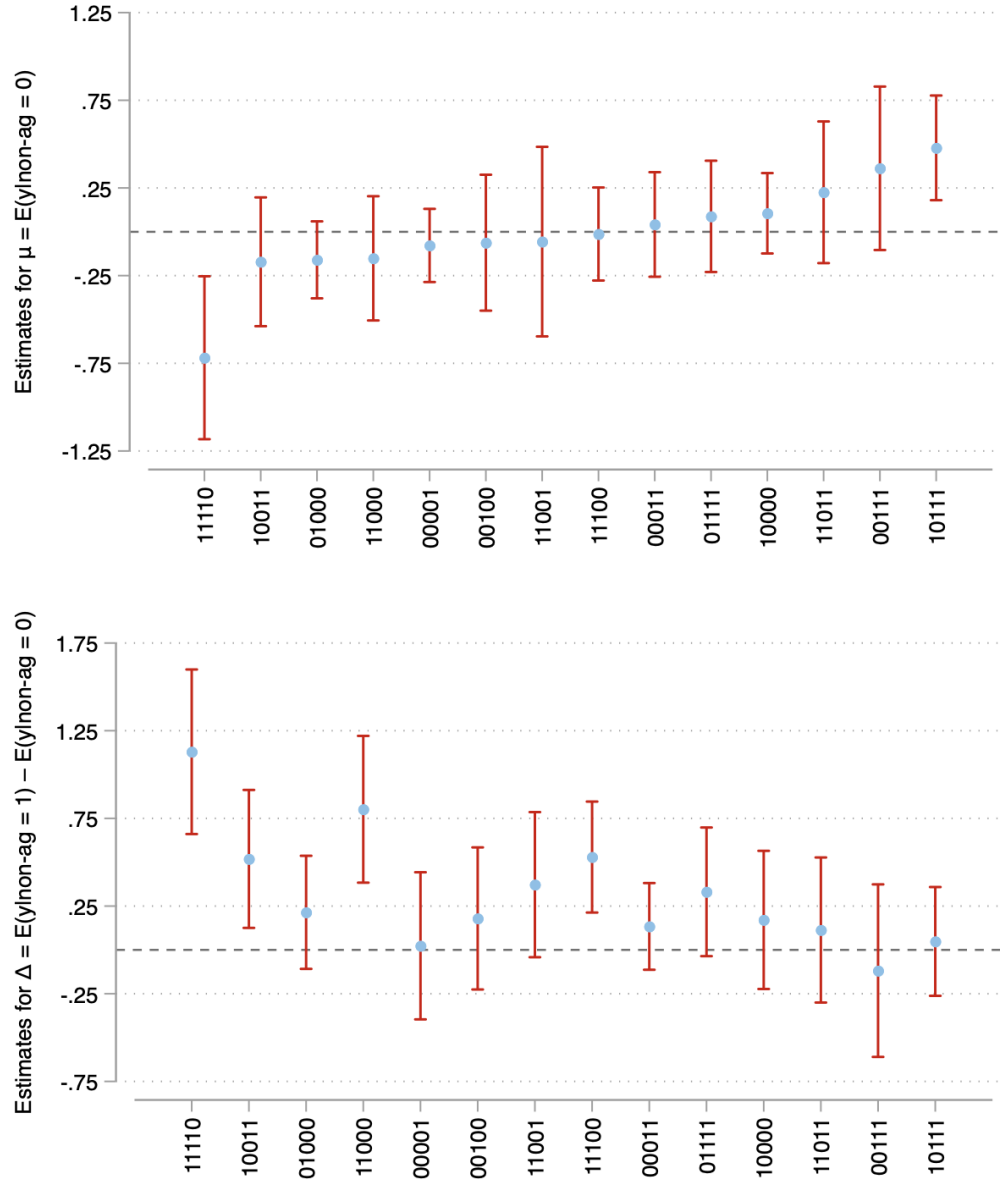
4.2 Results from GRC estimations

As mentioned in Section 2, we can estimate two different versions of the GRC. The first is the unrestricted version, which does not impose a linear decomposition of unobserved abilities. This version can flexibly estimate heterogeneous returns to nonagricultural employment (see equation (5) in our model). Specifically, it estimates the base income gain from working in agriculture for all trajectory types except one (the “always non-agriculture” type) and the incremental income gain from working in non-agriculture for all switcher types.

With five periods, we have $2^5 = 32$ trajectory types (or 30, if we remove the two switcher types). Some of these trajectory types have very few individuals in the sample, sometimes just one. Moreover, many types do not reflect a particular group of interest in public policy interventions. Therefore, instead of presenting the estimated nonagricultural premia for all trajectory types in a regression table like we did for the OLS results, we show estimates for selected types in a figure. This has the added benefit of highlighting two features of the heterogeneity in the returns to nonagricultural employment in our setting. First, the figure illustrates the considerable heterogeneity we have in both the base returns to working in agriculture and in the incremental returns to working in non-agriculture. Second, it provides visual support for the LCA assumption shown in equation (10), showing a negative relationship between the base returns in agriculture and the incremental returns in non-agriculture.

Figure 2 shows the results from unrestricted GRC estimations for selected trajectory types (all those with ten or more individuals in the data). For conciseness, we show only the results for the specification with both baseline covariates and time fixed effects (equivalent to column three in the regression tables).

Figure 2: Unrestricted GRC estimates of the returns in log income from nonagricultural employment for selected switcher types



Note: The blue dots are point estimates for μ and Δ obtained from estimating the unrestricted GRC using the specification shown in column three in all tables (baseline controls and time fixed effects are included as controls). The red bars show 95% confidence intervals. The dashed gray line is set at zero on the y-axis. Point estimates for μ are normalized such that the estimate for the “Never” type is zero. Selected switchers types are those with ten or more individuals in the sample. Labels in the x-axis show each type’s trajectory of nonagricultural employment in which 1 represents nonagricultural employment, and 0 represents agricultural employment, with the first period (IFLS wave) at the bottom and the last at the top. The types are ordered according to the magnitude of their point estimates for μ .

The top half of the figure shows estimates for the base income return to working in agriculture (the parameters μ_d in our model). The blue dots represent point estimates, and the red bars show 95% confidence intervals. We normalize the results so that the estimate for the non-switcher type $d = \{0, 0, 0, 0, 0\}$ (the “never non-agriculture” type) is zero. We then order the types along the x-axis from the smallest to the largest point estimate. We maintain this same order along the x-axis in the bottom half of the figure, in which we show the estimates for the incremental income return to working in non-agriculture (the parameters Δ_d in our model).

The figure suggests an inverse relationship between base returns in agriculture and incremental returns to switching to nonagricultural employment in our data. For any two pairs of trajectory types, the type with the lowest base return generally has the highest incremental return. This pattern is consistent with a negative extrapolation slope (the parameter ϕ in the restricted GRC model). Indeed, we find a negative sign for ϕ across specifications when estimating the restricted GRC model and imposing the linear extrapolation to identify the returns to nonagricultural employment for the two non-switcher types.

Table 4 shows the results for the restricted GRC estimations. For conciseness, we report only the estimates for the nonagricultural premium for the two non-switcher types, the “never” and the “always” types (respectively, $d = \{0, \dots, 0\}$ and $d = \{1, \dots, 1\}$). We also report the estimate for the extrapolation slope, ϕ , and the J-statistic for the overidentification test in the GMM procedure, along with its p-value. The specifications used in columns one to four follow the same progression of controls used in the OLS regressions.

As indicated in Figure 2, the estimated coefficient for the extrapolation slope, which identifies the returns to non-switchers, is negative. Nonagricultural employment can be seen as a “pro-poor” technology in our context: the trajectory types with the lowest base returns to agricultural employment get the highest incremental returns when switching to non-agriculture. The results in Table 4 also show that the nonagricultural premium for the “never” type, the individuals who are never observed switching from agriculture to non-agriculture in the data, is 28–30 lp (or 20 lp if we consider the last specification with province fixed effects). These estimates are close to the range of estimates we get for most switcher types. These results indicate that marginal and inframarginal workers would see similar income gains if they moved from the agricultural to the nonagricultural sector in our setting.

The results for the “always” type, however, are less precise. In all specifications except the last, the estimates are not statistically significant and vary in magnitude and direction. In the last, the estimate is positive and significant (a premium of 35 lp). However, the extrapolation parameter is not precisely estimated in the last specification, which casts doubt on the estimates produced by it. Therefore, we focus our interpretations on the results in columns one to three and take the results for the “always” type as being not indistinguishable

Table 4: Restricted GRC estimates of the returns in log income from nonagricultural employment for Non-Switcher types

Dependent variable:	Log income			
	(1)	(2)	(3)	(4)
NonAg. Employment $\times$ Never type	0.3036 (0.0441)***	0.2818 (0.0395)***	0.3038 (0.0440)***	0.2029 (0.0655)***
NonAg. Employment $\times$ Always type	0.0381 (0.1472)	0.0662 (0.1196)	-0.1461 (0.1531)	0.3477 (0.0652)***
Extrapolation slope (ϕ)	-0.3653 (0.1405)***	-0.4503 (0.1262)***	-0.6518 (0.0791)***	0.5653 (0.4211)
Overidentification test (J-stat)	26.41	26.94	34.01	27.94
p-value	0.1912	0.1729	0.0361	0.1419
Observations	7,090	7,090	7,090	7,090
Individuals	1,418	1,418	1,418	1,418
Baseline Controls		Y	Y	Y
Time (wave) FE			Y	Y
Province FE				Y

Notes: Types are defined according to individual trajectories of nonagricultural employment. The Never type is an indicator equal to one for individuals who are not observed in the nonagricultural sector in any of the five IFLS waves. The Always type is an indicator equal to one for individuals who are observed in the nonagricultural sector in all five IFLS waves. Baseline controls are an indicator for born urban, female dummy, household size (nr. of members), age and its square, and years of education and its square, all measured in the baseline (first wave). nonagricultural Employment is an indicator equal to one if the individual's main job is in the nonagricultural sector. Robust standard errors in parentheses. Stars denote: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

from zero. Finally, we note that we cannot reject the overidentifying restriction in three of the four specifications. Only in column three is it rejected.

5 Conclusion

Existing estimates of productivity gaps suggest that rural-urban migration and sectoral change can provide important avenues for increasing labor productivity, income, and welfare in developing countries. However, these estimates say little about the potential returns for groups of individuals who are particularly relevant for the design and evaluation of possible policy interventions: the non-switchers, the individuals who are always (or never) observed

in rural areas or the agricultural sector and for whom returns may be considerably different from those of switchers.

Focusing on sectoral choices and returns in income, we provide estimates of the returns for non-switchers by modeling self-selection into non-agriculture in a multi-period Roy model, which incorporates worker heterogeneity in both absolute and comparative advantage. We then estimate a correlated random coefficient model that considers both types of heterogeneity. We also draw on recent developments in the literature on non-parametric panel data identification and employ a group-random coefficient estimator that explicitly tests the parametric assumptions that identify the returns to non-switchers.

We reconcile some of the divergent estimates in the literature. Our results from pooled OLS and panel regressions with individual fixed effects show a large (0.67 log points) agricultural/nonagricultural income gap, which is later reduced by the inclusion of different controls and individual fixed effects. Estimations for a sample restricted to switchers—the individuals effectively contributing to the identifying variation in specifications with individual fixed effects—yield a range of estimates of around 30 log points. The estimations of our correlated random coefficients model highlight heterogeneity in the returns to sectoral choice for different groups of switchers and set the stage for us to estimate returns for the two non-switcher types. We find returns for the “never” type close to the average returns for switchers (20–30 log points) and returns for the “always” type that are not statistically different from zero in general. We also find evidence suggesting that employment in the nonagricultural sector is a “pro-poor” technology: the trajectory types with the lowest base returns to agricultural employment get the highest incremental returns when switching to non-agriculture.

Our theoretical framework and empirical application could be extended in several ways. First, the analysis can focus on rural/urban migration instead of sectoral mobility as the main choice variable that affects an individual’s income. Second, we can look at outcomes different than total income, such as wages or consumption, which are reasonable proxies for productivity and well-being. Third, we can incorporate information on geographical mobility and the moving costs associated with it. All these dimensions of individual mobility—across sectors, rural/urban locations, and space—are interesting for academic and policy reasons, as are the different outcomes that proxy for individual well-being (wages, income, and consumption at the individual or household level). Nonetheless, because our primary goal is to present a new approach to estimating returns to non-switchers, we focus on the choice dimension and the outcome of interest shown to be most relevant in the context of Indonesia: sectoral mobility from the agricultural to the nonagricultural sector and its impact on an individual’s total income.

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